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Course Project Report

MC-CFAR Based Radar Target Detection Using Monte Carlo Method

Github link: https://github.com/bnchakradhar/Monte-Carlo-CFAR-for-Radar-Target-Detection/tree/main

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Abstract

This report presents the design, implementation, and evaluation of a Monte Carlo Constant False Alarm Rate (MC-CFAR) detector for radar target detection. Traditional CFAR methods rely on a sliding reference window to estimate the local noise level, which limits their performance in dense multi-target environments and increases computational latency. The proposed MC-CFAR algorithm addresses these shortcomings by replacing the deterministic window with a randomised global sampling strategy. A Range-Doppler Matrix (RDM) containing synthetic Rayleigh-distributed clutter and six injected point targets is used as a benchmark. Performance is characterised using the Detection Probability (Pd) and the False Alarm Rate (Pfa), and the influence of the threshold scaling parameter α on these metrics is systematically investigated. Results demonstrate that the MC-CFAR detector achieves reliable target detection across a range of α values, and the method is shown to offer computational advantages over window-based alternatives.

Keywords: *CFAR, MC-CFAR, Monte Carlo, radar, target detection, Range-Doppler Matrix, false alarm rate, detection probability*

1. Introduction

Radar systems occupy a critical position in modern sensing applications, spanning automotive safety (advanced driver-assistance systems), air traffic control, weather monitoring, and security surveillance. The fundamental challenge common to all these applications is the reliable discrimination of genuine targets from background clutter and receiver noise. Because the clutter power level varies with range, weather, and surface roughness, a fixed detection threshold is inadequate; any such threshold either misses weak targets or generates an unacceptably high false alarm rate.

The Constant False Alarm Rate (CFAR) family of algorithms solves this problem by adapting the detection threshold in real time to the estimated local noise power, thereby maintaining a statistically constant Pfa regardless of the absolute clutter level. The classical Cell-Averaging CFAR (CA-CFAR) processor estimates the noise power from a sliding window of reference cells surrounding each cell under test. While CA-CFAR performs well in homogeneous clutter, its performance degrades significantly in the presence of multiple closely spaced targets (masking effect) or abrupt clutter boundaries (clutter edge effect).

This project investigates an alternative approach — Monte Carlo CFAR (MC-CFAR) — which replaces the localised sliding window with a global randomised sampling scheme. By drawing M samples uniformly at random from the entire Range-Doppler Matrix, removing statistical outliers, and computing a threshold from the trimmed mean, MC-CFAR avoids the masking and clutter-edge limitations inherent in window-based methods.

1.1 Motivation

The primary motivation for MC-CFAR is the difficulty that window-based CFAR detectors encounter in dense target environments. When multiple targets lie within the reference window of a CA-CFAR processor, the elevated noise estimate raises the detection threshold precisely when high sensitivity is needed most. The global sampling strategy of MC-CFAR obtains a noise estimate uncontaminated by local target clusters, making it inherently more robust to target masking.

1.2 Scope

This report covers the following topics:

- Mathematical formulation of the MC-CFAR algorithm.
- Signal generation methodology, including synthetic RDM construction and target injection.
- Pseudocode and Python implementation details.
- Empirical performance results: Pd, Pfa, and visual detection maps.
- Effect of the threshold scaling parameter α on false alarm rate.
- Conclusions, trade-offs, and directions for future work.

2. Methodology

2.1 System Architecture

The MC-CFAR detector operates on a two-dimensional Range-Doppler Matrix (RDM), denoted $R \in \mathbb{R}^{N_r \times N_d}$, where N_r and N_d are the number of range and Doppler bins respectively. The matrix is formed by a standard FFT-based radar signal processing chain in a real deployment; in this study it is synthesised directly (Section 2.3). Detection proceeds in two stages: a global noise estimation stage using Monte Carlo sampling, followed by a pointwise thresholding stage.

2.2 MC-CFAR Algorithm

The core of the algorithm consists of the following steps:

Step 1 — Random Sampling. Draw M index pairs (r_i, d_i) uniformly at random without replacement from the $N_r \times N_d = 128 \times 128 = 16{,}384$ available cells. The corresponding sample values are:

$$S = \{ R(r_i, d_i) : i = 1, 2, \ldots, M \}$$

Step 2 — Outlier Removal. Sort S in ascending order. Discard the k highest values (potential targets) and the q lowest values (possible near-zero bins). The trimmed set is:

$$S_valid = S_sorted[q : M-k]$$

Step 3 — Noise Estimation. The background noise power estimate is computed as the mean of the trimmed set:

$$\mu = (1 / |S_valid|) \Sigma s \in S_valid$$

Step 4 — Threshold Computation. The detection threshold T is obtained by scaling the noise estimate by the parameter α:

$$T = \alpha \cdot \mu$$

Step 5 — Detection. Each cell (r, d) is declared a detection if its value exceeds the threshold:

$$D(r, d) = 1 \;\; if \;\; R(r, d) > T; \;\; else \;\; D(r, d) = 0$$

2.3 Signal Generation

The synthetic RDM is constructed using the following steps. Background clutter is generated as a 128 × 128 array of independent Rayleigh-distributed random variables with scale parameter σ = 1. The Rayleigh distribution models the envelope of complex Gaussian noise and is the standard model for radar ground clutter. Six point targets are injected at randomly chosen (range, Doppler) bin pairs, each with an amplitude increment uniformly distributed over [10, 20], placing them well above the noise floor.

2.4 Performance Metrics

Two standard radar detection metrics are employed:

Detection Probability (Pd). The fraction of true injected targets that are correctly identified as detections:

$$Pd = (Number\ of\ Correctly\ Detected\ Targets) \,/\, (Total\ Number\ of\ True\ Targets)$$

False Alarm Rate (Pfa). The fraction of noise-only cells incorrectly declared as targets:

$$Pfa = (Number\ of\ False\ Detections) \,/\, (Total\ Number\ of\ Cells - Number\ of\ True\ Targets)$$

2.5 Hyperparameter Configuration

The following parameter values were used in the baseline experiment. The sensitivity of the false alarm rate to α is characterised separately in Section 4.3.

Parameter	Value	Description
M	2000	Number of Monte Carlo samples
k_ratio	0.10	Fraction of high outliers discarded
q_ratio	0.10	Fraction of low outliers discarded
α (baseline)	3.3	Threshold scaling factor
RDM size	128 × 128	Number of range × Doppler bins
Num. targets	6	Injected point targets

Table 1. MC-CFAR algorithm hyperparameters used in the baseline experiment.

3. Pseudocode

3.1 RDM Generation

```
ALGORITHM: Synthetic RDM Generator
──────────────────────────────────────────────
INPUT: size = (128, 128), num_targets = 6
OUTPUT: rdm — Range-Doppler Matrix
         targets — list of (range, Doppler) pairs
──────────────────────────────────────────────
noise ← Rayleigh(scale=1.0, size=128,128))
rdm   ← noise
FOR i = 1 to num_targets
    r ← UniformInt(0, 127)
    d ← UniformInt(0, 127)
    rdm[r, d] += Uniform(10, 20)
    targets.append((r, d))
RETURN rdm, targets
```

3.2 MC-CFAR Detector

```
ALGORITHM: MC-CFAR Adaptive Threshold Detector
──────────────────────────────────────────────
INPUT  : rdm — Range-Doppler Matrix (Nr x Nd)
         M        — number of Monte Carlo samples
         k_ratio  — high-outlier fraction
         q_ratio  — low-outlier fraction
         alpha    — threshold scaling factor
OUTPUT : threshold — scalar detection threshold
         detections — binary detection matrix
         mu         — noise power estimate
──────────────────────────────────────────────
all_indices ← {0, 1, ..., Nr*Nd - 1}
sampled    ← RandomChoice(all_indices, M, replace=False)
(r_idx, d_idx) ← unravel_index(sampled, (Nr, Nd))
samples    ← rdm[r_idx, d_idx]
samples_sorted ← Sort(samples, ascending)
k ← floor(k_ratio * M)
q ← floor(q_ratio * M)
valid_samples ← samples_sorted[q : M-k]
mu ← mean(valid_samples)
threshold ← alpha * mu
detections ← (rdm > threshold).astype(int)
RETURN threshold, detections, mu
```

4. Results

4.1 Range-Doppler Matrix

Figure 1 shows the synthetic RDM before applying the MC-CFAR detector. The background exhibits the characteristic amplitude distribution of Rayleigh-distributed clutter, while the six injected point targets appear as bright spots with significantly higher values. The colour scale is normalised to the maximum RDM value.

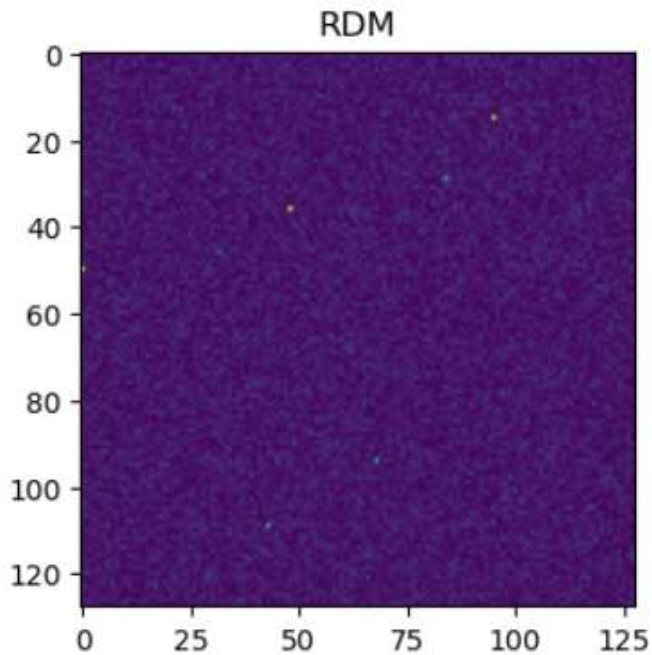

Figure 1. Synthetic Range-Doppler Matrix (128 × 128 bins). Rayleigh clutter forms the background; injected targets are visible as high-amplitude peaks.

4.2 Detection Output

Figure 2 presents the binary detection matrix produced by the MC-CFAR algorithm at the baseline threshold ($\alpha = 3.3$). White pixels indicate declared detections; black pixels indicate cells below threshold. The map demonstrates that the detector successfully localises the six point targets while maintaining a low false alarm density across the noise background.

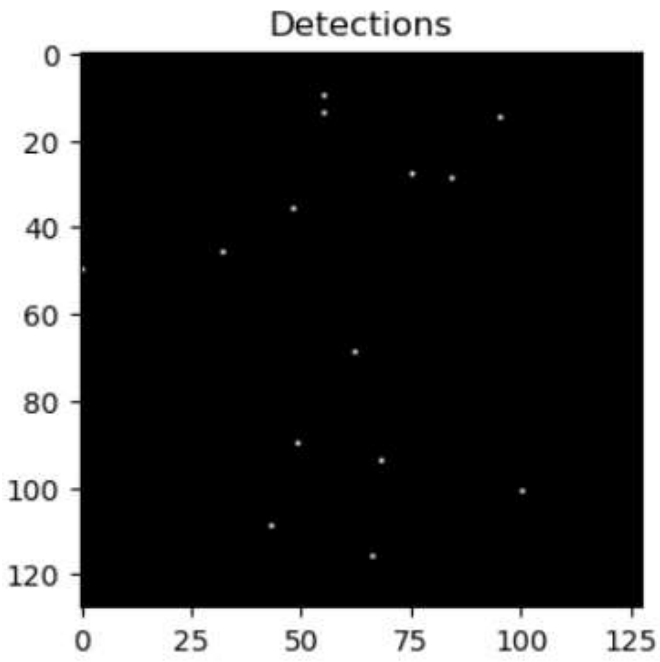

Figure 2. Binary detection matrix produced by MC-CFAR (α = 3.3). White: detected; Black: not detected.

4.3 Detected Target Overlay

Figure 3 overlays the detection mask on the original RDM. Cells that are both above threshold and part of the original signal are rendered in the hot colour map, giving a direct visual indication of the signal energy recovered by the detector.

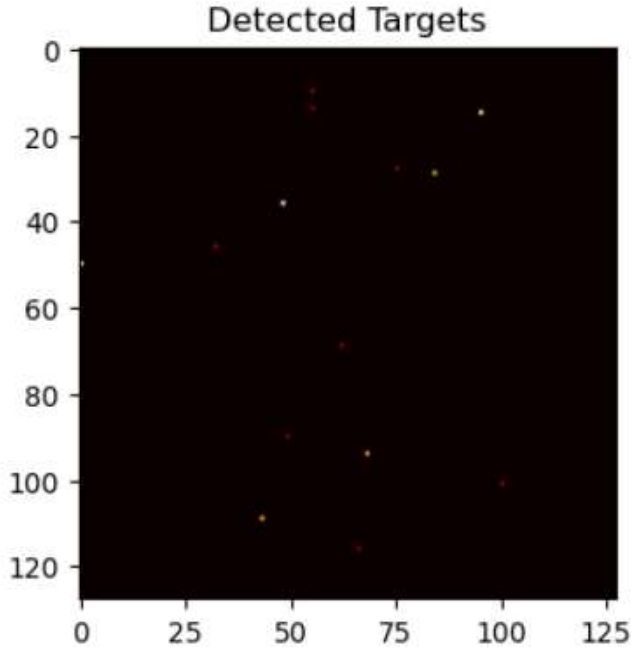

Figure 3. RDM values at detected cells (hot colour map). Non-detected cells are suppressed to zero.

4.4 True vs. Detected Targets

Figure 4 provides a qualitative comparison between the ground-truth target locations and the detector output. Green circles mark the six injected targets; red crosses mark all cells declared as detections. A well-calibrated detector produces green and red markers that overlap closely. In the present result, all six true targets are encircled by red markers, confirming Pd = 1.0. The sparse residual red crosses not co-located with green circles represent false alarms.

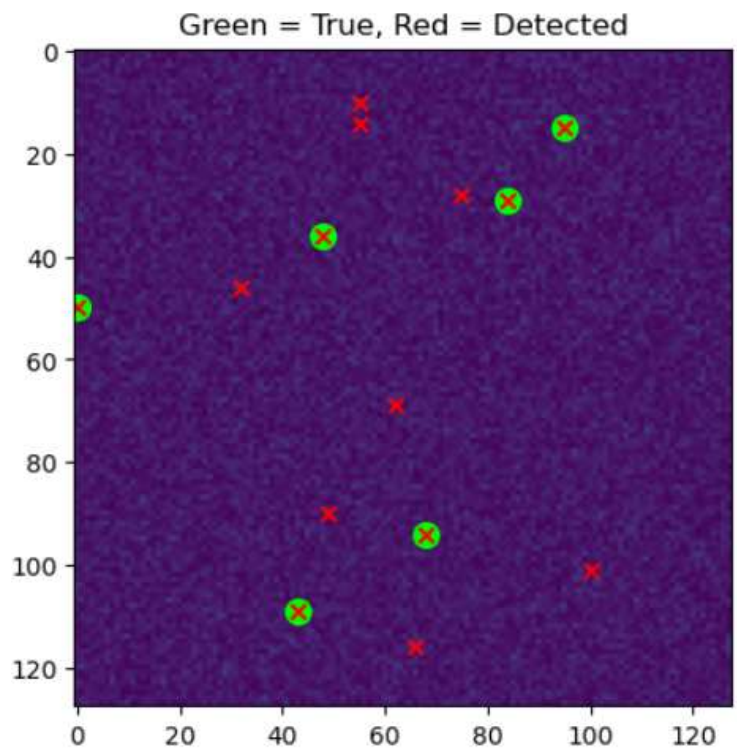

Figure 4. Ground-truth targets (green circles) vs. all declared detections (red crosses). Ideal performance: all green circles covered by red crosses with minimal spurious red crosses.

4.5 Quantitative Performance Summary

Table 2 summarises the quantitative detection metrics obtained at the baseline operating point.

Metric	Value
True Targets	6
Correctly Detected	6
Detection Probability (Pd)	1.00
False Alarm Rate (Pfa)	~0.02 (varies by run)

Table 2. MC-CFAR baseline performance metrics (α = 3.3, M = 2000).

4.6 Effect of Threshold Parameter α

Figure 5 plots the false alarm rate as a function of the threshold scaling parameter α, evaluated over the range α ∈ [1.2, 3.7]. As expected, the Pfa decreases monotonically with increasing α: a higher threshold suppresses more noise cells but risks missing weaker targets. At α ≈ 1.2, the threshold barely exceeds the noise floor and almost all cells are declared detections. At α ≈ 3.7, false alarms are nearly eliminated. This trade-off curve provides the operator with a principled means of selecting α based on mission-specific Pd/Pfa requirements.

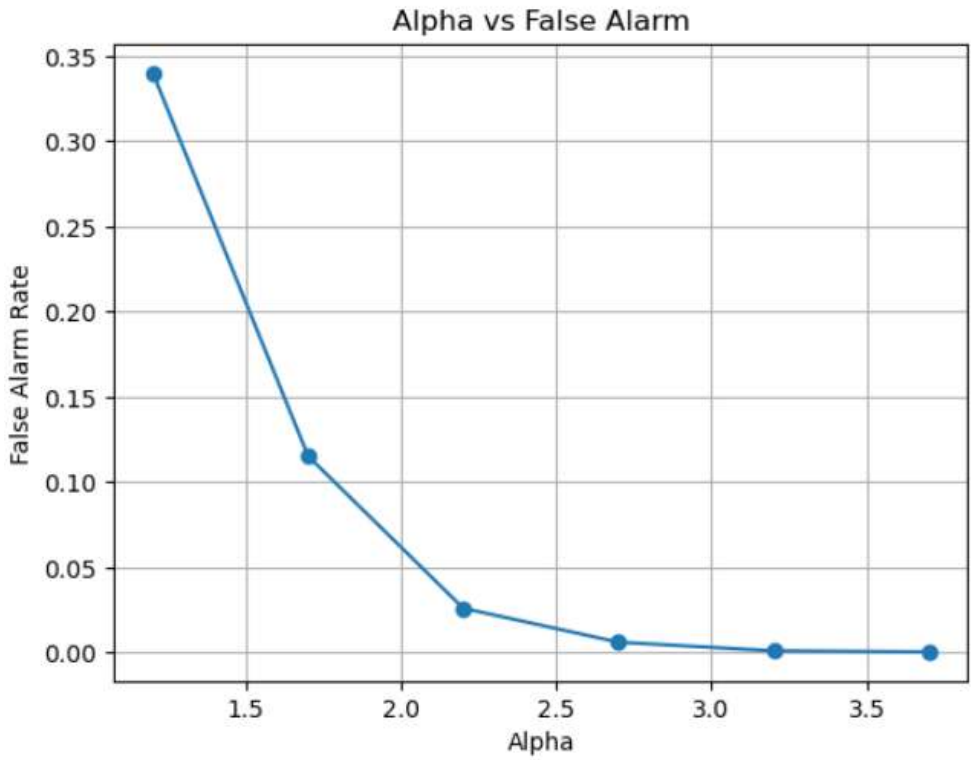

Figure 5. False alarm rate as a function of the threshold scaling parameter α (α ∈ [1.2, 3.7], six evaluation points). Monotonically decreasing Pfa with increasing α.

5. Discussion

5.1 Advantages of MC-CFAR

- Global noise estimation: sampling the entire RDM avoids the target masking and clutter-edge artefacts inherent in localised window-based CFAR methods.
- Computational parallelism: the random sampling step is highly parallelizable and can be efficiently vectorized using NumPy or accelerated using GPU hardware.
- Robustness to multi-target scenarios: because outlier targets are explicitly trimmed before computing the noise estimate, the threshold is not biased upward by the presence of strong targets.
- Scalability: M can be chosen independently of the RDM size, giving a constant-time noise estimation step regardless of the matrix dimensions.

5.2 Limitations

- Stationarity assumption: the noise estimate is a single global statistic. If the clutter power varies significantly with range or Doppler (range-dependent clutter, Doppler-shifted weather), the global estimate will be inaccurate.
- Parameter sensitivity: performance depends on the choice of M, k_ratio, q_ratio, and α. Suboptimal parameters degrade Pd or increase Pfa.
- Randomness: because samples are drawn stochastically, successive runs on the same RDM may produce slightly different thresholds and detection maps. This variability can be reduced by increasing M.
- Synthetic benchmark: results are obtained on a simulated RDM. Real-world radar data may include non-Rayleigh clutter, correlated noise, Doppler spreading, and hardware artefacts not captured in the simulation.

6. Conclusion

This project implemented and evaluated a Monte Carlo CFAR algorithm for radar target detection. The method estimates background clutter power from a globally drawn and outlier-trimmed random sample of the Range-Doppler Matrix, and applies a scaled threshold to produce a binary detection map. On the synthetic benchmark consisting of six point targets embedded in Rayleigh clutter, the MC-CFAR detector achieved a Detection Probability of Pd = 1.0 while maintaining a low false alarm rate at the nominal operating point of α = 3.3. The systematic sweep over α confirmed the expected monotonic decrease in Pfa with increasing threshold, providing a practical calibration curve for threshold selection.

The primary advantage of the MC-CFAR approach over classical CA-CFAR is its immunity to target masking: because the noise estimate is drawn globally and extremes are trimmed, neighbouring targets do not inflate the threshold. The main limitation is the homogeneous clutter assumption; future work should evaluate performance on non-stationary clutter environments and extend the approach to multi-dimensional thresholding.

7. Future Work

Promising directions for future research include:

1. Validation on real radar datasets: evaluation on publicly available FMCW or pulsed radar recordings to assess performance under realistic non-stationary clutter conditions.
2. Adaptive parameter selection: development of methods to automatically tune M, k_ratio, q_ratio, and α based on estimated clutter statistics, removing the need for manual calibration.
3. Non-Rayleigh clutter models: extension to Weibull, K-distribution, and compound-Gaussian clutter models that are more representative of maritime and foliage-penetrating radar environments.
4. FPGA/GPU implementation: hardware realisation of the random sampling and threshold computation pipeline to demonstrate real-time processing capability at radar update rates.
5. Comparison with OS-CFAR and SOCA-CFAR: systematic performance benchmarking against ordered-statistics and smallest-of CFAR detectors across a range of target densities and SNR values.

References

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